MTH 203 : Lecture 8

Definition:

(1) If $f: D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$ $(n>1)$ is a real (scalar) valued function of several variables then it is also called a Scalar field.

(2) If $f: D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$ $(n>1,\ m>1)$ is a vector valued function of several variables then it is also called a Vector field.

f can be written as

$$\vec{f}(\vec{x}) = \left(f_1(\vec{x}), f_2(\vec{x}), \ldots, f_m(\vec{x})\right)$$

where the component functions $f_i: D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$ is a scalar field for $i = 1, 2, \ldots, m$.

- We will try to generalize the concept of Total Derivative for vector field:

Let $\vec{f}: D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be a vector valued function of n variables (i.e. $\vec{f}$ is a vector field).

Let $\vec{f}(\vec{x}) = \left(f_1(\vec{x}), f_2(\vec{x}), \ldots, f_m(\vec{x})\right)$

where $f_i\ D \longrightarrow \mathbb{R}$ (component functions) for $1 \leq i \leq m$

Differentiability:

Let $\vec{f} : D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$, $\vec{x}_0 \in \text{int}(D)$

We say $\vec{f}$ is differentiable at $\vec{x}_0$ (or $\vec{f}$ has a total derivative at $\vec{x}_0$)

if $\exists$ a linear transformation

$$T_{\vec{x}_0} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

such that

$$\lim_{\vec{h} \to 0} \frac{\overbrace{\| \vec{f}(\vec{x}_0 + \vec{h}) - \vec{f}(\vec{x}_0) - T_{\vec{x}_0}(\vec{h}) \|}^{\in \mathbb{R}^m}}{\|h\|} = 0$$

- When the limit exists, we set $\vec{f}'(x_0) = T_{\vec{x}_0}$ and call it the Total Derivative of $\vec{f}$ at $\vec{x}_0$

Total Derivative in terms of Partial Derivatives:

Theorem: Let $\vec{f} : D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be a vector valued function of n independent variables $x_1, x_2, \dots, x_n$.

Let $\vec{f} = (f_1, f_2, \dots, f_m)$

So, $f_i : D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$ for $1 \le i \le m$ denote the component functions of $\vec{f}$.

If $\vec{f}$ is differentiable at $\vec{x}_0$ then for all $1 \le i \le m$ and $1 \le j \le n$, $\frac{\partial f_i}{\partial x_j}(\vec{x}_0)$ exists and $\vec{f}\,'(\vec{x}_0) = T_{\vec{x}_0} = \left[\frac{\partial f_i}{\partial x_j}(\vec{x}_0) \right]_{m \times n}$

Proof: Exercise

Jacobian Matrix:

Definition: Let $\vec{f} : D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be a vector valued function of n independent variables $x_1, x_2, \dots x_n$ and let $f_i : D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$ for $1 \le i \le m$ denote the component functions of $\vec{f}$

Let $\vec{f}$ be differentiable at $\vec{x}_0$

Let $T_{\vec{x}_0} = \vec{f}\,'(x_0)$ be the total derivative of $\vec{f}$ at $\vec{x}_0$.

Then

- $T_{\vec{x}_0}$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

- The $m \times n$ matrix of the linear transformation $T_{\vec{x}_0}$ with respect to the standard basis of $\mathbb{R}^n$ and $\mathbb{R}^m$ is called the Jacobian Matrix of the function $\vec{f}$ at the point $\vec{x}_0$.

Ex ①: Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ be given by

$$f(x, y) = \sin x$$

Compute $f'((2,3))$

$$f'((x,y)) = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right] = [\cos x, 0]$$

$$\Rightarrow \boxed{f'((2,3)) = [\cos 2, 0]}$$

Ex ②: Let $\vec{f} : \mathbb{R}^2 \to \mathbb{R}^3$ be defined by

$$\vec{f}(x,y) = (\sin x \cos y,\ \sin x \sin y,\ \cos x \cos y)$$

Determine the Jacobian matrix of $\vec{f}$ at any arbitrary point $(x,y) \in \mathbb{R}^2$.

- $\vec{f}\,'((x,y))$ (It will be a 3×2 matrix)

$$= \begin{bmatrix} \frac{\partial f_1(x,y)}{\partial x} & \frac{\partial f_1(x,y)}{\partial y} \\ \frac{\partial f_2(x,y)}{\partial x} & \frac{\partial f_2(x,y)}{\partial y} \\ \frac{\partial f_3(x,y)}{\partial x} & \frac{\partial f_3(x,y)}{\partial y} \end{bmatrix}$$

$$= \begin{bmatrix} \cos x \cos y & -\sin x \sin y \\ \cos x \sin y & \sin x \cos y \\ -\sin x \cos y & -\cos x \sin y \end{bmatrix}$$

Ex: Let us assume that the range is 1-dimensional.

So, $f : D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}$, $\vec{x_0} \in \text{int}(D)$,

Let f be differentiable at $\vec{x_0}$.

Then $f'(\vec{x_0})$ is a $1 \times n$ matrix given by:

$$\begin{bmatrix} \frac{\partial f}{\partial x_1}(\vec{x_0}) & \frac{\partial f}{\partial x_2}(\vec{x_0}) & \cdots & \frac{\partial f}{\partial x_n}(\vec{x_0}) \end{bmatrix}_{1 \times n}$$

- The Jacobian matrix of f at $\vec{x_0}$ is a row matrix and is called "The gradient of f at $\vec{x_0}$".

It is denoted by

$$\nabla f \Big|_{\vec{x_0}} \quad \text{or} \quad \nabla f(\vec{x_0})$$

<u>Note:</u>

- $\nabla f(\vec{x_0}) = \left[\frac{\partial f}{\partial x_1}(\vec{x_0}), \cdots \frac{\partial f}{\partial x_n}(\vec{x_0}) \right]$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}$ and is called the <u>gradient of f at $\vec{x_0}$.</u>
- However $\nabla f = \left[\frac{\partial f}{\partial x_1}, \cdots, \frac{\partial f}{\partial x_n} \right]$ is called <u>the gradient of f</u> or <u>simply 'del f'.</u>

Note that ∇f is <u>not</u> a <u>linear transformation from $\mathbb{R}^n$ to $\mathbb{R}$.</u>

Actually $\nabla f : E \subseteq \mathbb{R}^n \longrightarrow M_{1\times n}(\mathbb{R})$

where $E \subseteq D$ is the set of all points in D where the partial derivatives $\frac{\partial f}{\partial x_j}$ exists for every $1 \leq j \leq n$

Ex: $f(x,y) = x^2 y$

$$\underbrace{\nabla f}_{\text{a function from } \mathbb{R}^2 \text{ to } M_{1,2}(\mathbb{R})} = \left[\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \right] = \left[2xy \quad x^2 \right] \rightarrow \text{gradient of } f$$

$$\underbrace{\nabla f \Big|_{(1,2)}}_{\text{a linear transformation from } \mathbb{R}^2 \text{ to } \mathbb{R}.} = \left[2\times 1 \times 2 , \; 1^2 \right] = [4, 1] \rightarrow \text{gradient of } f \text{ at } (1,2)$$

Ex: The Domain is one dimensional.

Let $\vec{f} : D \subseteq \mathbb{R} \longrightarrow \mathbb{R}^m$, $x_0 \in \text{int}(D)$

be a differentiable at x_0

Then $\vec{f}\,'(x_0)$ is the $m \times 1$ matrix

$$\begin{bmatrix} \frac{\partial f_1(x_0)}{\partial x} \\ \frac{\partial f_2(x_0)}{\partial x} \\ \vdots \\ \frac{\partial f_m(x_0)}{\partial x} \end{bmatrix}_{m \times 1} = \begin{bmatrix} \frac{\partial f_i(x_0)}{\partial x} \end{bmatrix}_{m \times 1}$$

In this case we don't use the '∂' notation and use ordinary derivative sign.

So, the Jacobian matrix is a column matrix and is denoted by

$$\vec{f}'(x_0) = \begin{bmatrix} \frac{df_1(x_0)}{dx} \\ \frac{df_2(x_0)}{dx} \\ \vdots \\ \frac{df_m(x_0)}{dx} \end{bmatrix} = \frac{d\vec{f}(x_0)}{dx}$$

Directional Derivative as a Dot Product

- Let $f : D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}$ and let f be a differentiable function in a region containing the point (x_0, y_0)

Then $$D_{\vec{u}} f \Big|_{(x_0, y_0)} = \nabla f \Big|_{(x_0, y_0)} \cdot \vec{u}$$

Ex:

Find the directional derivative of $f(x, y) = 2x^2 + y^2$ at $(-1, 1)$ (Note: Done earlier directly) along $(3i - 4j)$

$$\vec{u} = \left(\frac{3}{\sqrt{3^2 + (-4)^2}}, \frac{-4}{\sqrt{3^2 + (-4)^2}} \right) = \left(\frac{3}{5}, \frac{-4}{5} \right)$$

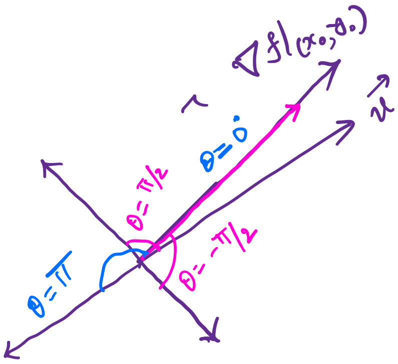
$$\nabla f = [4x, 2y]$$

$$\nabla f \Big|_{(-1, 1)} = [-4, 2]$$

$$\nabla f \Big|_{(-1,1)} \cdot \vec{u} = [-4, 2] \begin{bmatrix} \frac{3}{5} \\ -\frac{4}{5} \end{bmatrix}$$

$$= -\frac{12}{5} - \frac{8}{5} = -\frac{20}{5} = \boxed{-4}.$$

Proposition: $$D_{\vec{u}} f\Big|_{(x_0,y_0)} = \nabla f\Big|_{(x_0,y_0)} \cdot \vec{u}$$



$$= \left| \nabla f\Big|_{(x_0,y_0)} \right| \, |\vec{u}| \cos\theta$$

$$= \left| \nabla f\Big|_{(x_0,y_0)} \right| \cos\theta$$

- when $\theta = 0$, then $\vec{u}$ is parallel to ∇f and thus $D_{\vec{u}} f\big|_{(x_0,y_0)}$ assumes the maximum value when we choose $\vec{u}$ in the direction of ∇f. In this case $D_{\vec{u}} f\big|_{(x_0,y_0)} = \left|\nabla f\big|_{(x_0,y_0)}\right|$

Thus

- At each point in its domain, f increases most rapidly in the direction of ∇f. $(\theta = 0^\circ)$

and

- f decreases most rapidly in the direction of $-\nabla f$ $(\theta = \pi)$

In this case
$$D_{\vec{u}} f\Big|_{(x_0,y_0)} = -\left| \nabla f\Big|_{(x_0,y_0)} \right|$$

- Any direction $\vec{u}$ orthogonal to the gradient $\nabla f \,(\neq 0)$ is a direction of zero change because

  $\theta = \pi/2$ will imply $\cos \pi/2 = 0$

  Thus in this case $D_{\vec{u}} f \Big|_{(x_0, y_0)} = 0$

Ex: $f(x, y) = x^2 + xy$

Find the directional derivative of f at $(1, 2)$ in the direction of $(1, 1)$.

- First calculate it using the above formula

- $\vec{u} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$

$\nabla f = [\,2x + y \,, \; x\,]$

$\nabla f \Big|_{(1,2)} = [\,2 \times 1 + 2, \; 1\,] = [4, 1]$

So, $D_{\vec{u}} f \Big|_{(1,2)} = \nabla f \Big|_{(1,2)} \cdot \vec{u}$

$= [4, 1] \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

$$= 4 \times \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{5}{\sqrt{2}} = \boxed{\frac{5\sqrt{2}}{2}}$$

Using the definition:

$$D_{\vec{u}} f \Big|_{(1,2)} = \lim_{h \to 0} \frac{f\left((1,2) + h\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)\right) - f(1,2)}{h}$$

$$= \lim_{h \to 0} \frac{f\left(1 + \frac{h}{\sqrt{2}}, 2 + \frac{h}{\sqrt{2}}\right) - f(1,2)}{h}$$

$$= \lim_{h \to 0} \frac{\left(1 + \frac{h}{\sqrt{2}}\right)^2 + \left(1 + \frac{h}{\sqrt{2}}\right)\left(2 + \frac{h}{\sqrt{2}}\right) - \left(1^2 + 1 \times 2\right)}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{1} + \frac{h^2}{2} + 2\frac{h}{\sqrt{2}} + \cancel{2} + 2\frac{h}{\sqrt{2}} + \frac{h^2}{2} + \frac{h}{\sqrt{2}} \; -3}{h}$$

$$= \lim_{h \to 0} \frac{h^2 + 5\frac{h}{\sqrt{2}}}{h}$$

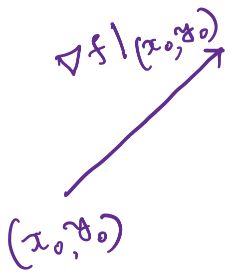
$$= \lim_{h \to 0} \frac{\cancel{h}\left(h + \frac{5}{\sqrt{2}}\right)}{\cancel{h}} = \lim_{h \to 0} \left(h + \frac{5}{\sqrt{2}}\right)$$

$$= 0 + \frac{5}{\sqrt{2}} = \frac{5}{\sqrt{2}} = \boxed{\frac{5\sqrt{2}}{2}}$$

<u>Ex:</u> $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$

Find the direction in which f increases (decreases) most rapidly at $(1, 1)$.

What are the direction of zero changes of f at $(1, 1)$?



$$\nabla f = \left[\frac{2x}{2}, \frac{2y}{2}\right] = [x, y]$$

$$\nabla f\Big|_{(1,1)} = [1, 1] \quad (\text{or } \vec{i} + \vec{j})$$

Hence $\vec{u} = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$ $\left(\text{or, } \frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j}\right)$

So, f will increase most rapidly in the direction $\frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j}$

f will decrease most rapidly in the direction $-\frac{1}{\sqrt{2}}\vec{i} - \frac{1}{\sqrt{2}}\vec{j}$

<u>Directions of zero changes</u>

$$-\frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j} \quad \text{and} \quad \frac{1}{\sqrt{2}}\vec{i} - \frac{1}{\sqrt{2}}\vec{j}$$

<u>Note:</u> If $\vec{v} = v_1\vec{i} + v_2\vec{j}$ then $\vec{u} \cdot \vec{v} = 0 \Rightarrow v_1 + v_2 = 0 \Rightarrow v_1 = -v_2$

The gradient or ∇-operator:

Let $f : D \subseteq \mathbb{R}^2 \longrightarrow \mathbb{R}$

$(x_0, y_0) \in \mathrm{int}(D)$

$$\nabla f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} \quad \left(\text{or } \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right)\right)$$

(of course to be evaluated at (x_0, y_0) to obtain $\nabla f|_{(x_0,y_0)}$)

Symbolically ∇ is a Differential operator and formally written as: $\boxed{\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y}}$ or $\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right)$

- For $f : D \subseteq \mathbb{R}^3 \longrightarrow \mathbb{R}$

$$\nabla = \vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}$$

$$\text{or } \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$$

Algebra of gradients:

- $\nabla(cf) = c\nabla f$ for any constant c
- $\nabla(f \pm g) = \nabla f \pm \nabla g$

(check!)

- $\nabla(fg) = f\nabla g + g\nabla f$
- $\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2}$

Ex: The directional derivative of $f(x,y)$ at $(1,2)$ in the direction of $\vec{i}+\vec{j}$ is $2\sqrt{2}$ and in the direction of $-2\vec{j}$ is -3. What is the directional derivative in the direction of $-\vec{i}-2\vec{j}$

The unit vector along $\vec{i}+\vec{j}$ is $\vec{u} = \frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j}$

$$\Rightarrow D_{\vec{u}} f\Big|_{(1,2)} = \left[\frac{\partial f}{\partial x}(1,2), \frac{\partial f}{\partial y}(1,2)\right]\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2}\end{bmatrix} = \frac{\partial f(1,2)}{\partial x}\frac{1}{\sqrt{2}} + \frac{\partial f(1,2)}{\partial y}\frac{1}{\sqrt{2}}$$
$$= 2\sqrt{2}$$

$$\Rightarrow f_x(1,2) + f_y(1,2) = 4 \quad \text{.......} (1)$$

Now the unit vector along $-2\vec{j}$ is $\vec{v} = -\vec{j}$

$$\Rightarrow D_{\vec{v}} f\Big|_{(1,2)} = f_x(1,2)\times 0 + f_y(1,2)(-1) = -3$$
$$\Rightarrow f_y(1,2) = 3 \quad \text{......} (2)$$

Substituting (2) in (1) we get $f_x(1,2) = 1$ and $f_y(1,2) = 3$

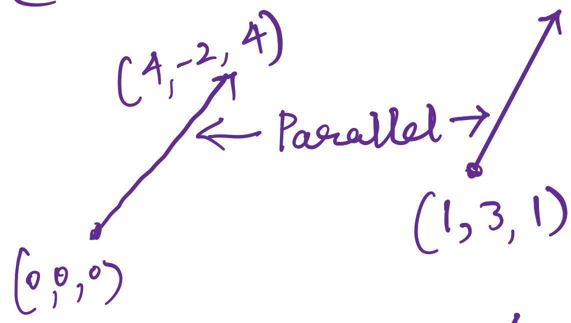
Now in the direction $-\vec{i}-2\vec{j}$, the unit vector $\vec{w} = \left(-\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}\right)$

Then $D_{\vec{w}} f\Big|_{(1,2)} = \nabla f\Big|_{(1,2)} \cdot \left(-\frac{1}{\sqrt{5}}\vec{i} - \frac{2}{\sqrt{5}}\vec{j}\right)$

$$= (\vec{i}+3\vec{j})\cdot\left(-\frac{1}{\sqrt{5}}\vec{i} - \frac{2}{\sqrt{5}}\vec{j}\right) = -\frac{1}{\sqrt{5}} - \frac{6}{\sqrt{5}} = \boxed{\frac{-7}{\sqrt{5}}}$$

Ex: $f(x,y,z) = x^2 - y + z^2$

Find the directional derivative at $(1,3,1)$ in the direction of $4\vec{i} - 2\vec{j} + 4\vec{k}$



Note that $\vec{u} = \left(\dfrac{4}{\sqrt{4^2+(-2)^2+4^2}}, \dfrac{-2}{\sqrt{4^2+(-2)^2+4}}, \dfrac{4}{\sqrt{4^2+(-2)^2+4^2}} \right)$

$$= \frac{4}{6}\vec{i} - \frac{2}{6}\vec{j} + \frac{4}{6}\vec{k}$$

$$= \frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} + \frac{2}{3}\vec{k}$$

$$\nabla f = [2x, -1, 2z]$$

$$\nabla f\big|_{(1,3,1)} = [2, -1, 2]$$

Hence $D_{\vec{u}} f \big|_{(1,3,1)} = [2, -1, 2] \cdot \begin{bmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \end{bmatrix}$

$$= \frac{4}{3} + \frac{1}{3} + \frac{4}{3} = \frac{9}{3} = \boxed{3}$$

Note: Calculate the same using the definition of directional derivative.